

Class 1: Probability & Statistics



Disclaimer

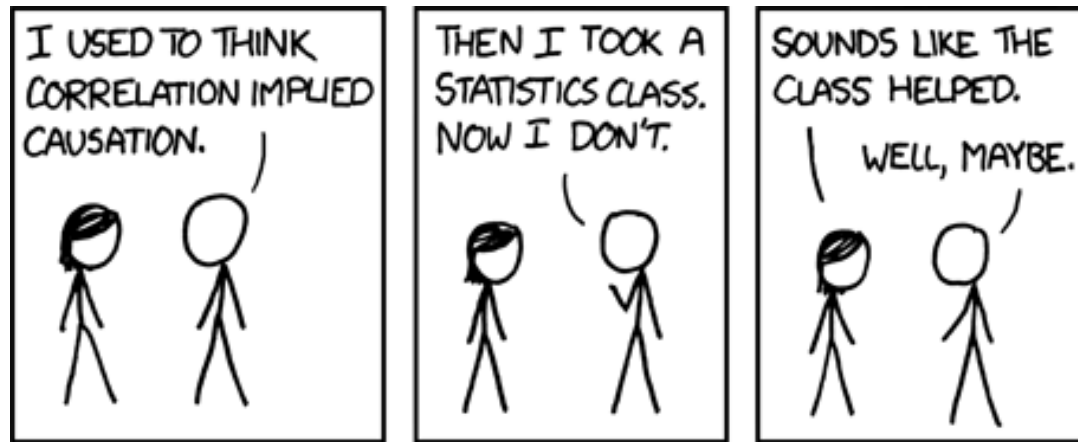
In your PhD studies you might be using some detailed statistical techniques in your topic

In this course, we'll focus on a **broad toolbox of approaches** that should be relevant in the physical sciences



(clicked image of toolbox)

Quick intro to statistics course



- **Week 1:** Probability and statistics
- **Week 2:** Correlation testing
- **Week 3:** Model fitting
- **Week 4:** Regression
- **Week 5:** Error estimates
- **Week 6:** Bayesian methods

What to do in the course each week?

- Visit the **course module** on the PhD reviews Canvas site
- View each week's **video** (~30 min) – slides are also available
- Work through each week's **Jupyter notebook**, which includes coding examples and activities to try
- Come to our **weekly class** (Tuesdays 3-4pm in AMDC801) to chat about the concepts!
- To **complete the course**, please send me your code for the final activities in each Jupyter notebook
- Whole course is designed for 20 hours' study

Topics for today

At the end of this class you should be able to ...

- ... determine **summary statistics** for datasets and their errors
- ... optimally **combine** data
- ... apply **probability distributions** for Gaussian, Binomial and Poisson statistics
- ... compare the **Frequentist and Bayesian** frameworks for statistical analysis
- ... solve statistical problems using **Monte Carlo** techniques

Summary statistics: worked example

A statistic is a quantity which **summarises** our data



- We have $N = 5$ measurements of a quantity: $(7.4 \pm 2.0, 6.5 \pm 1.1, 4.2 \pm 2.5)$. What is the **optimal estimate** of this quantity and the **error in that estimate**?

$$\hat{y} = \frac{\sum_{i=1}^N x_i / \sigma_i^2}{\sum_{i=1}^N 1 / \sigma_i^2}$$

$$\frac{1}{\text{Var}(\hat{y})} = \sum_{i=1}^N \frac{1}{\sigma_i^2}$$

Summary statistics: worked example

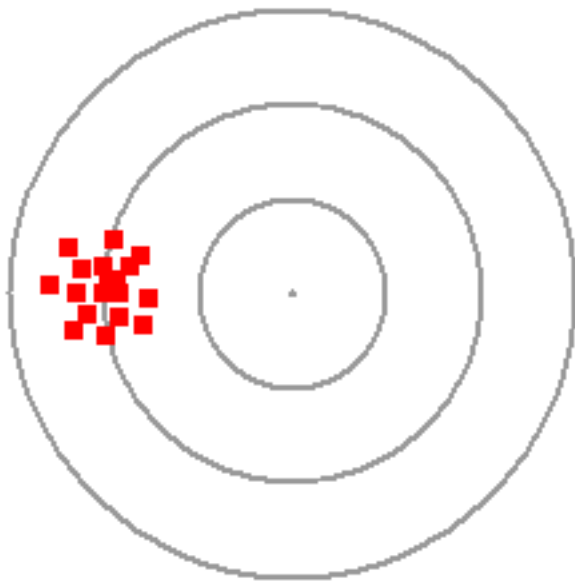
A statistic is a quantity which **summarises** our data



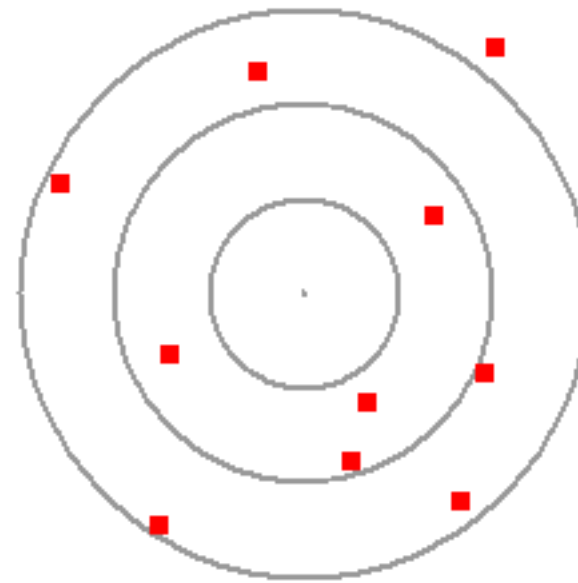
- We have $N = 5$ measurements of a quantity: $(7.4 \pm 2.0, 6.5 \pm 1.1, 4.2 \pm 2.5)$. What is the **optimal estimate** of this quantity and the **error in that estimate**?
- A further measurement 3.0 ± 0.2 is added. How should our estimate change?
- Why should we be cautious about adding this measurement?

Summary statistics: worked example

Optimally weighting data is only useful when we are dealing with **random errors** rather than **systematic errors**! What is the difference?



Systematic Error



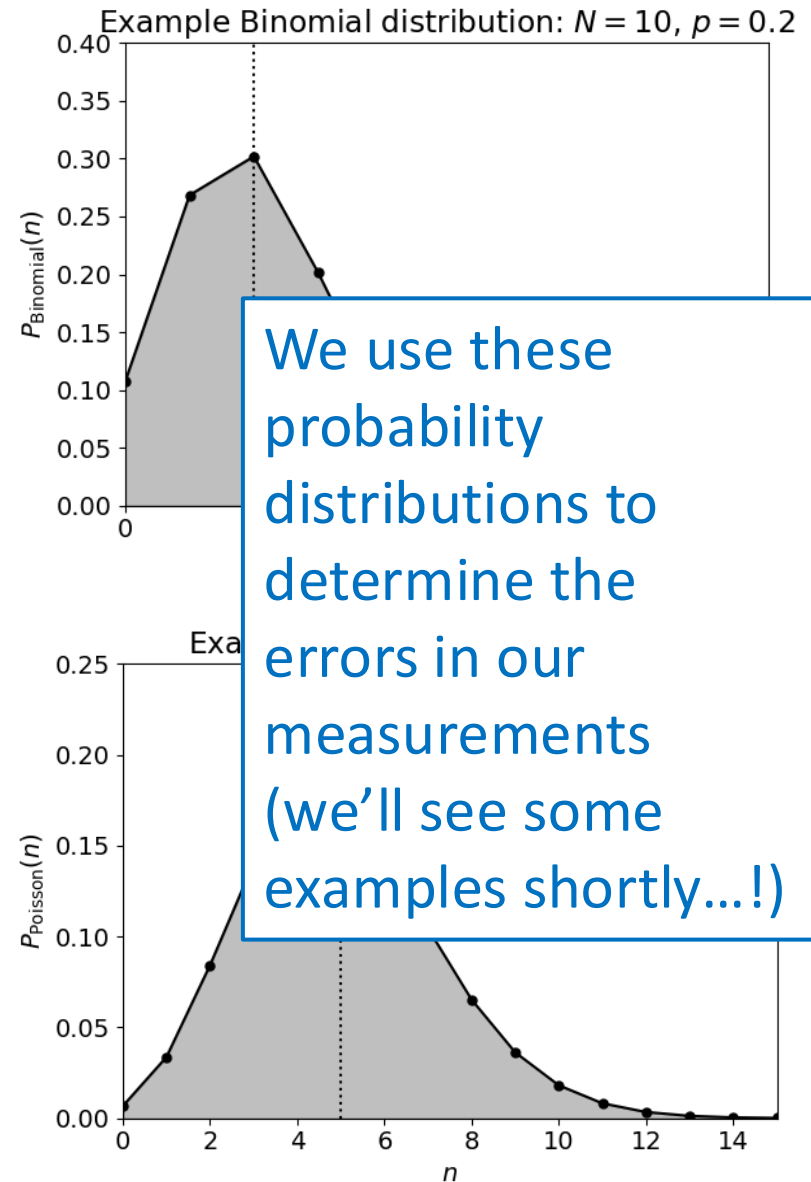
Random Error

Let's remind ourselves of some distributions...

What type of process is described by a binomial distribution?

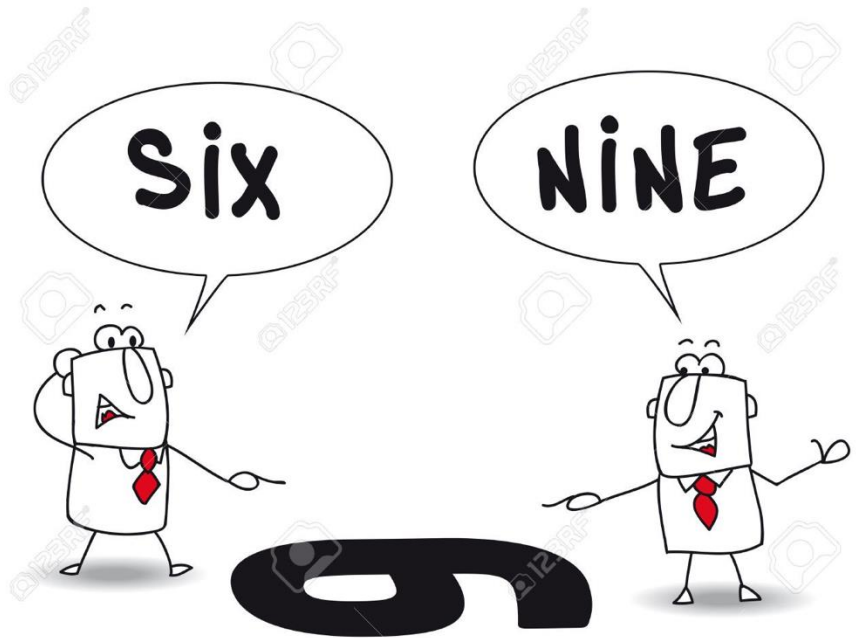
What type of process is described by a Poisson distribution?

Why is a Gaussian distribution so common in statistics?



Frequentist and Bayesian frameworks

- When discussing statistics, we will often hear about “**Frequentist**” or “**Bayesian**” methods. In the next few slides we’ll discuss what this means.
- Neither framework is “right” or “wrong”, as such
- *As usual with statistics, it comes down to the question we want to answer ...*



Frequentist and Bayesian frameworks

- The concept of **conditional probability** is central to understanding Bayesian statistics
- $P(A|B)$ means “**the probability of A on the condition that B has occurred**”
- Adding conditions makes a huge difference to evaluating probabilities!
- On a randomly-chosen day in CAS, $P(\text{free pizza}) \sim 0.2$
- $P(\text{free pizza}|\text{Monday}) \sim 1$, $P(\text{free pizza}|\text{Tuesday}) \sim 0$

Frequentist and Bayesian frameworks

- **Frequentist statistics** assign probabilities to a measurement, i.e. they determine $P(\text{data}|\text{model})$
- We are defining probability by imaging a series of **hypothetical experiments**, repeatedly sampling the population (which have not actually taken place)
- *Philosophy of science: we attempt to “rule out” or falsify models, if $P(\text{data}|\text{model})$ is too small*



Assuming these dice are unbiased, what is the probability of rolling different values?

Frequentist and Bayesian frameworks

- **Bayesian statistics** assign probabilities to a model, i.e. they give us tools for calculating $P(\text{model}|\text{data})$
- We **update the model probabilities** in the light of each new dataset (rather than imagining many hypothetical experiments)
- *Philosophy of science: we do not “rule out” models, just determine their relative probabilities*



Assuming I roll a particular spread of different values, what is the probability of the dice being unbiased?

Frequentist and Bayesian frameworks

- The important formula for relating conditional probabilities is **Bayes' theorem**:

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$



(Obligatory portrait of the Reverend Bayes!)

- Re-writing Bayes' theorem for science:

$$P(\text{model}|\text{data}) = \frac{P(\text{data}|\text{model}) P(\text{model})}{P(\text{data})}$$

Usually can be ignored because it is absorbed into the normalisation

Usually set to a constant over the range we are testing

Worked example

- *I observe 100 galaxies, 30 of which are AGN. What is the best estimate of the AGN fraction and its error?*
- There are 2 possible outcomes (“AGN” or “not an AGN”) so the **binomial distribution** applies
- **Frequentist:** Estimate AGN fraction $p = \frac{N_{AGN}}{N_{total}} = \frac{30}{100} = 0.3$
- Estimate the error in N_{AGN} as the standard deviation in the binomial distribution $= \sqrt{N_{total} p(1-p)} = \sqrt{100 \times 0.3 \times 0.7} = 4.6$, so error in $p = \frac{4.6}{100} = 0.046$
- Answer: $p = 0.3 \pm 0.046$

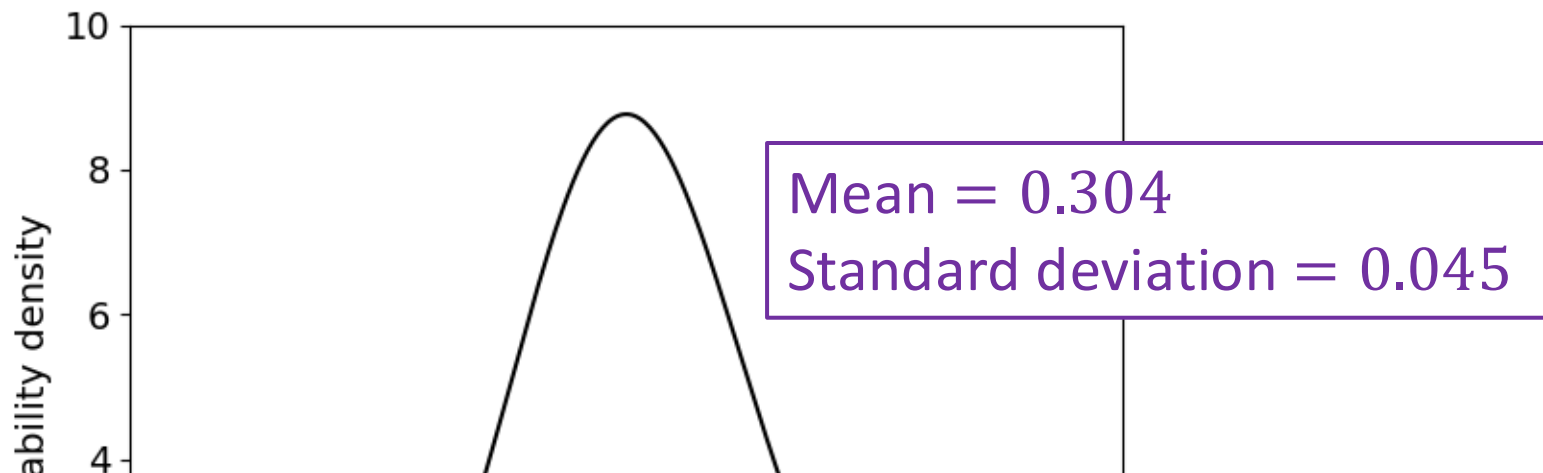
There is a flaw in this logic!

Worked example

- *I observe 100 galaxies, 30 of which are AGN. What is the best estimate of the AGN fraction and its error?*
- **Bayesian:** Use Bayes' theorem $P(p|D) \propto P(D|p) P(p)$
- $P(p|D)$ is the probability distribution of p given the data D , the quantity we aim to determine
- $P(D|p)$ is the probability of the data for a given value of p , which is given by the Binomial distribution as $P_{\text{Binomial}}(n = 30|N = 100, p)$
- $P(p)$ is the prior in p , which we take as a uniform distribution between $p = 0$ and $p = 1$
- Determining $P(p|D)$ and normalising we obtain ...

Worked example

- I observe 100 galaxies, 30 of which are AGN. What is the best estimate of the AGN fraction and its error?*



The Bayesian method gives us the probability distribution of the quantity we are trying to determine – i.e., $P(p|D)$

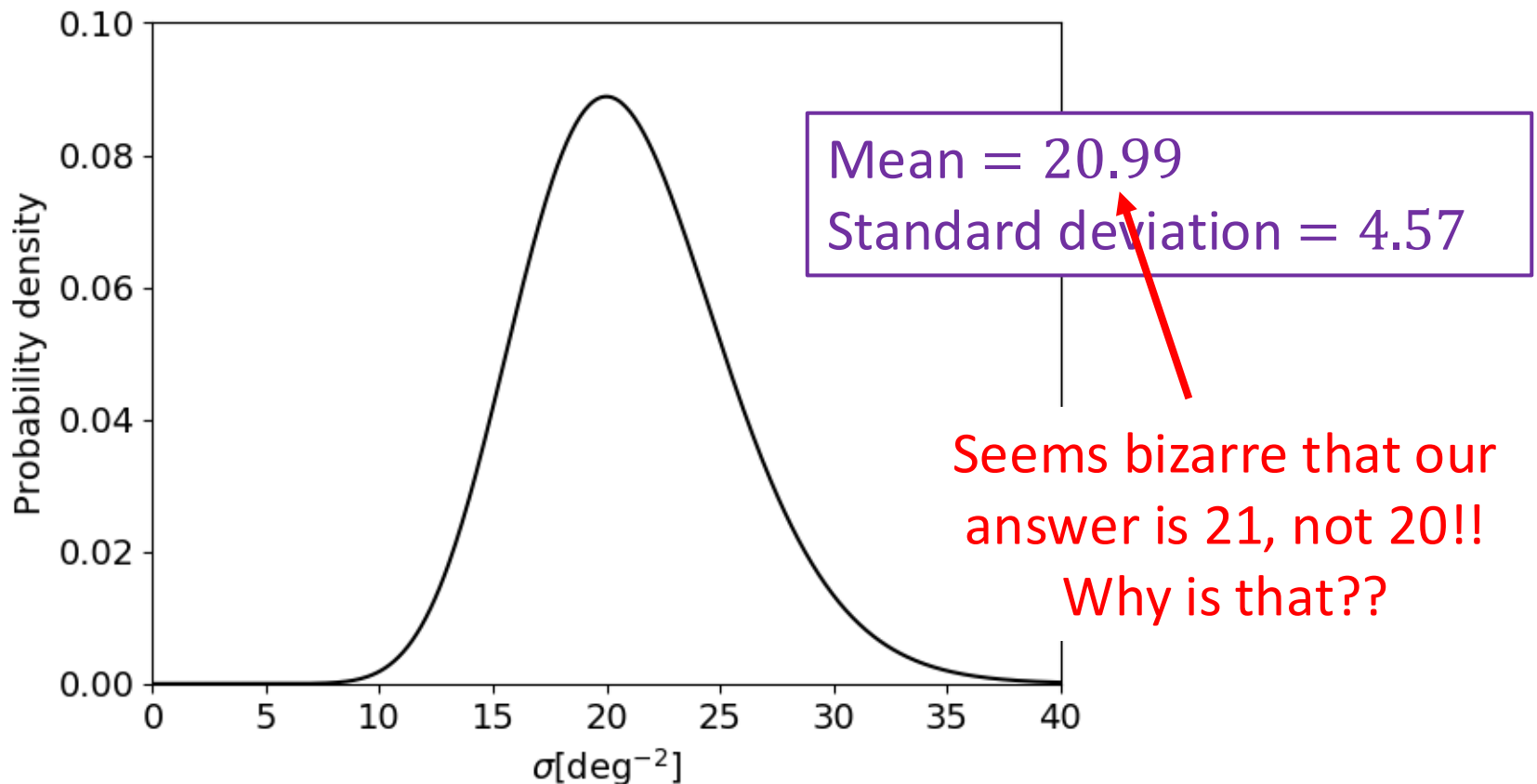
In the Frequentist method, we assumed $p = 0.3$, without considering it could take other values, i.e. we only considered $P(D|p)$

Activity: statistical distributions

- Solve the following question using both a Frequentist and Bayesian approach:
- *A survey of area $A = 1 \text{ deg}^2$ finds $N = 20$ quasars. What is the number of quasars per square degree, σ , and its error?*
- This is a counting problem so the **Poisson distribution** applies
- **Frequentist logic:** Density = $N \pm \sqrt{N} = 20 \pm 4.5$
- **Bayesian logic:** $P(\sigma|N = 20) \propto P_{\text{Poisson}}(N = 20|\sigma) P(\sigma)$
where prior $P(\sigma)$ is a constant
- Evaluating $P(\sigma|N = 20)$ and normalising we obtain...

Activity: statistical distributions

- A survey of area $A = 1 \text{ deg}^2$ finds $N = 20$ quasars. What is the number of quasars per square degree, σ ?*



Activity: statistical distributions

```
# Frequentist approach for quasar density  
n = 20  
print('Density =', n, '+/-' , np.sqrt(n))
```

Density = 20 +/- 4.47213595499958

```
# Bayesian approach for quasar density  
n = 20  
xmin, xmax, nx = 0., 40., 400  
dx = (xmax-xmin)/nx  
x = np.linspace(xmin+0.5*dx, xmax-0.5*dx, nx)  
prob = np.empty(nx)  
for i in range(nx):  
    prob[i] = stats.poisson.pmf(n, x[i])  
ymax = 0.1  
prob /= np.sum(prob)*dx  
x1 = np.sum(x*prob*dx)  
x2 = np.sum((x**2)*prob*dx)  
mu = x1  
sig = np.sqrt(x2 - x1**2)  
print('Density =', mu, '+/-' , sig)  
plt.plot(x, prob, color='black')  
plt.xlim(xmin, xmax)  
plt.ylim(0., ymax)  
plt.xlabel(r'$\sigma$ [{\rm deg}]$^{-2}$')
```

Density = 20.99231794460161 +/- 4.565774505744054

Monte Carlo simulations

- A **Monte Carlo simulation** is a computer model of an experiment in which many random realizations of the results are created and analysed like the real data

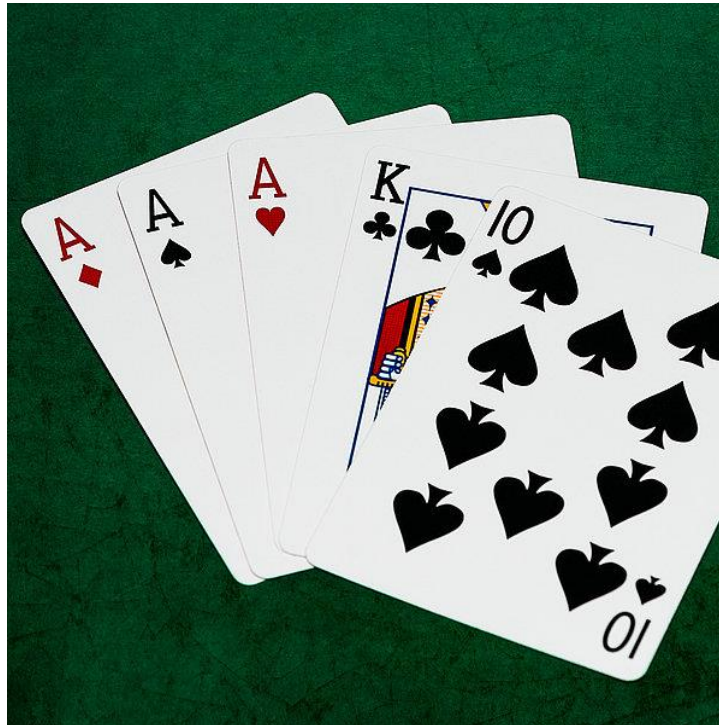


Monte Carlo simulations

- A **Monte Carlo simulation** is a computer model of an experiment in which many random realizations of the results are created and analysed like the real data
- ***This is the most useful statistical tool you'll learn!***
- It allows us to determine the statistics of a problem **without** any analytic calculations (if we can model it)
- Statistical errors can be obtained from the **distribution of fitted parameters** over the realizations
- Systematic errors can be explored by comparing the mean fitted parameters to their **known input values**

Activity: Monte Carlo methods

- Solve the following problem by Monte Carlo methods: *I'm dealt 5 playing cards from a normal deck (i.e. 13 different values in 4 suits). What is the probability of obtaining "three of a kind" (i.e. 3 of my 5 cards having the same value?)*



Activity: Monte Carlo methods

- My attempt!

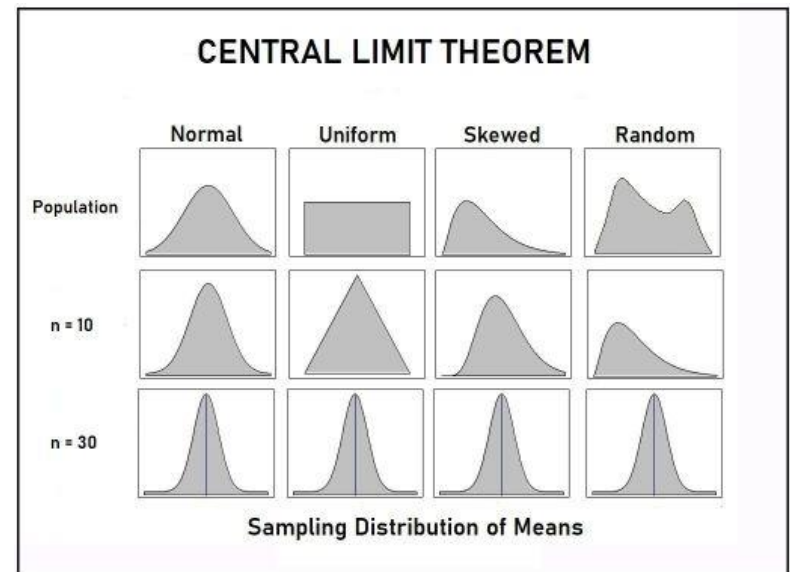
```
: # Monte Carlo simulation of poker!
nreal = 100000
deck = np.repeat(np.arange(13),4)
isave = 0
for ireal in range(nreal):
    ind = np.random.choice(52,5,replace=False)
    cards = deck[ind]
    freq = np.bincount(cards)
    if (np.isin(3,freq) and (not np.isin(2,freq))):
        isave += 1
print('Probability =',float(isave)/float(nreal))
```

Probability = 0.02127

Central limit theorem

What is the central limit theorem?

The **central limit theorem** says that if draw variables many times from any probability distribution, and average the result, the averages will follow a Gaussian distribution!



Why is this useful?

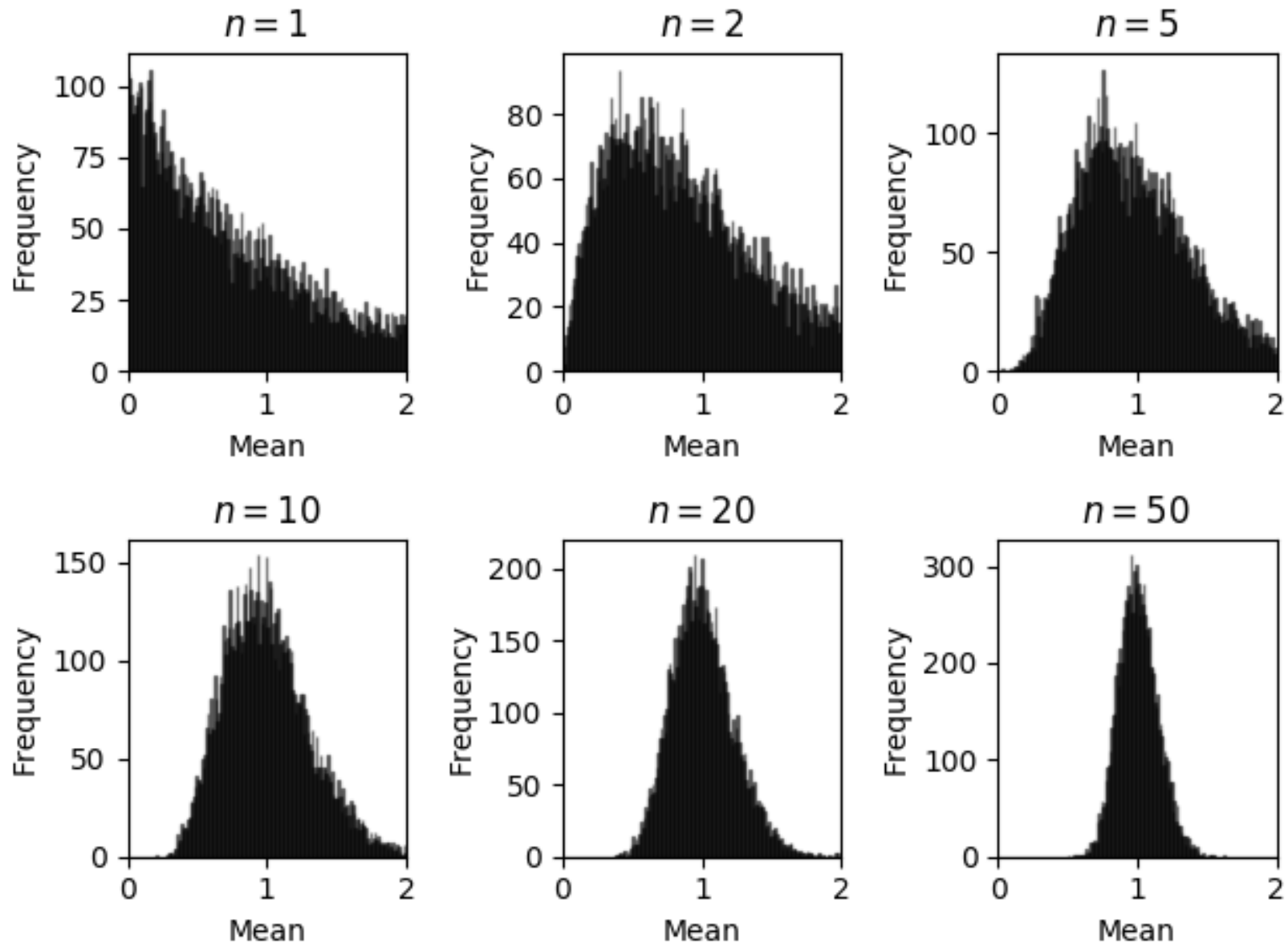
We intuitively understand Gaussian distributions!

Activity: central limit theorem

- Another example of a Monte Carlo simulation ...
- Write a code that draws n values of x from an exponential distribution $P(x) \propto e^{-x}$ (where $0 < x < \infty$), and computes their arithmetic mean μ . Repeat this process m times, and plot the probability distribution of μ across the m realisations. Run this experiment for values $n = 1, 2, 5, 10, 20, 50$.
- Hint: to do a single draw, select a uniform random number y in the range $0 < y < 1$, then $x = -\ln y$ [why does this work?]

The point being, this is very far from a Gaussian distribution – yet the averages of many draws follow a Gaussian!

Activity: central limit theorem



Next week: Correlation Testing

